

ME420 Mechanical Engineering Individual Research Project

PROJECT SELECTION FORM

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Name of Supervisor/s : Dr. D.H.S. Maithripala

Title of the project : Wearable-Based Worker Fatigue Monitoring System

Scope of the project (what exactly is to be done in the project):

- Design and develop a low-cost wearable band for construction workers(I plan to use an ESP32 as the MCU, but because it is not very small, making a compact wearable band may be difficult).
- Integrate PPG, IMU, and infrared skin temperature sensors into the wearable.
- Collect physiological, movement, and temperature data during construction-related tasks.
- Develop a data processing and machine learning model to estimate fatigue levels.
- Validate the accuracy and reliability of the fatigue estimation system.

Expected final outcome (what is the expected end result):

- A functional wearable fatigue monitoring band suitable for construction environments.
- A validated fatigue prediction model that estimates workers' fatigue levels in real time.

Important milestones hoping to be achieved (which should be measurable) during the project:

- Research gap clearly identified.
- Wearable band hardware designed and assembled.
- Sensor data acquisition system tested and calibrated.
- Feature extraction and data preprocessing completed.
- Fatigue prediction model developed and trained.
- Model accuracy evaluated and validated.
- Final report and presentation completed.

Signature of student

Signature of supervisor

Name of Supervisor: Dr. D.H.S. Maithripala

Date: 23/12/2025